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The Long-Term Economic Consequences of the African Slave Trade: A Critical Replication and Extension of Nunn (2008)

ABSTRACT
In light of Nathan Nunns' 2008 perennial work on the causal relationship between slave exports (1400-1900) and the contemporary economic development of African nations, this replication exercise has successfully reproduced and verified the robust negative relationship between historical slave exports and GDP per capita. Furthermore, our instrumental-variable estimation suggests that the effect is approximately twice the OLS estimate due to attenuation bias. The replication exercise has extended Nunn's (2008) work by estimating the mediation of slavery's effect on GDP per capita across ethic fractionalization and state capacity. Our findings suggest a statistically significant direct effect of slavery on concurrent GDP and no indication of mediation. Nonetheless, the analysis has acknowledged the effect of slavery exports on ethnic fractionalization, but not state development.

Keywords: Mediation Analysis, Instrumental Variables, Slave Trade, Colonial Legacies

INTRODUCTION

Between 1400 and 1900, Africa experienced one of the largest systems of the slave trade, in which over 18 million people were exported in the span of 500 years from Atlantic transfers and the Indian Ocean (Nunn, 2008, p. 139). One particular consequence of this endeavor has been the long-run effects on economic development, as Nunn (2008) demonstrates in his perennial research on slave trade and development: the intensity of slave exports relative to country size is associated with a strong negative statistical relationship with contemporary GDP per capita. The time span may be broad, but the consequences for socio-economic development are severe, with Manning (1990) estimating that populations in Africa would have reached only half of projected levels in the absence of the slave trade. The contribution of Nunn (2008) presents a case in which historical and colonial legacies have an innate effect on development trajectories today, combining contemporary debates on path dependencies and the role of geography in shaping development.

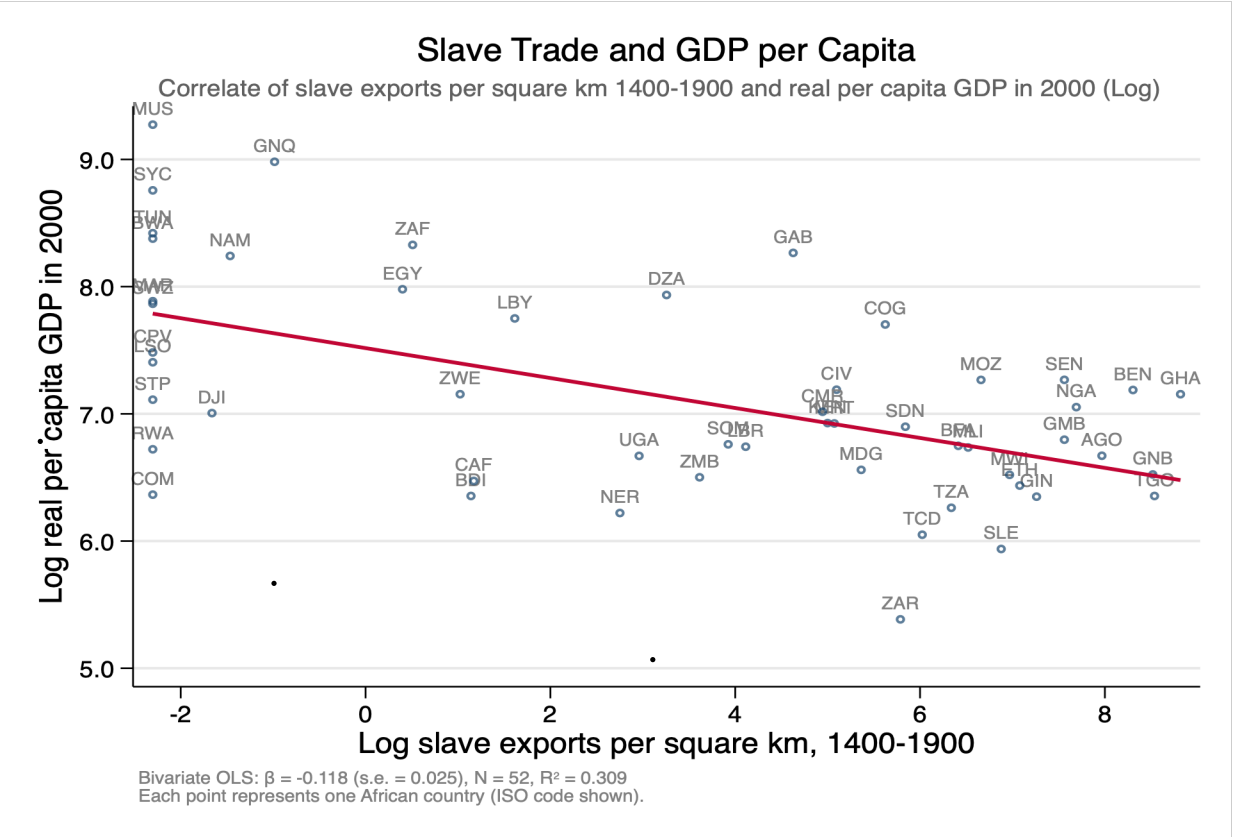

Figure 1. Correlates of Slave Exports per Square Meter 14-1900 and GDP per Capita in 2000 (Log). Data Source: Maddison, 2003.

For our replication exercise, Figure 1 represents the relationship that will be tested in our replication of the main OLS estimation of Nunn (2008), of which our dependent variable is Log real per capita GDP in 2000 (Maddison, 2003) against our main independent variable, ln(exports/area), with key geographic, institutional, and natural resource controls. (Table 1).

REPLICATION: OLS and IV estimation

Table 1. Replication of Nunn (2008) Table III - OLS Estimation

Dependent Variable Log Real Per Capita GDP in 2000, ln *y*	(1)	(2)	(3)	(4)	(5)	(6)
ln(exports/area)	-0.112***	-0.076**	-0.108***	-0.085**	-0.103***	-0.128***
	(0.024)	(0.029)	(0.037)	(0.035)	(0.034)	(0.034)
Distance from equator		0.016	-0.005	0.019	0.023	0.006
		(0.017)	(0.020)	(0.018)	(0.017)	(0.016)
Longitude		0.000	-0.007	-0.004	-0.004	-0.009
		(0.005)	(0.006)	(0.006)	(0.005)	(0.005)
Lowest monthly rainfall		-0.001	0.008	0.000	-0.001	-0.002
		(0.007)	(0.008)	(0.007)	(0.006)	(0.007)
Avg max humidity		0.009	0.008	0.009	0.015	0.013
		(0.012)	(0.012)	(0.012)	(0.011)	(0.010)
Avg min temperature		-0.019	-0.039	-0.005	-0.015	-0.037
		(0.028)	(0.028)	(0.027)	(0.026)	(0.023)
ln(coastline/area)		0.085**	0.092**	0.095**	0.082**	0.083**
		(0.039)	(0.042)	(0.042)	(0.040)	(0.037)
Island indicator				-0.398	-0.150	
				(0.529)	(0.516)	
Percent Islamic				-0.008***	-0.006*	-0.003
				(0.003)	(0.003)	(0.003)
French legal origin				0.755	0.643	-0.141
				(0.503)	(0.470)	(0.734)
North Africa indicator				0.382	-0.304	
				(0.484)	(0.517)	
ln(gold prod/pop)					0.011	0.014
					(0.017)	(0.015)
ln(oil prod/pop)					0.078***	0.088***
					(0.027)	(0.023)
ln(diamond prod/pop)					-0.039	-0.048
					(0.043)	(0.041)
Colonizer fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Restricted sample	No	No	Yes	No	No	Yes
Observations	52	52	42	52	52	42
R^2	0.50	0.60	0.63	0.71	0.77	0.80

*Standard Errors in Parentheses. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.10$*

Table 1 confirms the paper's key findings, particularly the negative statistical relationship between slave exports and GDP per capita, with minor discrepancies or noise in the replication. Controlling for geography reduces the effect size of export slaves/area, while additional institutional and resource controls increase the regression's explanatory power, and the slave export coefficient remains robust.

While providing a plausible statistical phenomenon, a challenge arises from potential endogeneity in the relationships, such as the selection of initially prosperous regions into the slave trade, which could bias OLS estimates. Despite this, Nunn shows the opposite relationship: more developed areas exported more slaves (p. 158, Figure IV), which would entail a positive selection bias towards zero. There may be additional measurement error due to limited observations in the OLS estimations, potentially leading to undersampling of interior slaves. Nunn addresses such concerns, however, by controlling for distance to slave markets as a foundational instrument.

Table 2. Replication of Nunn (2008) Table IV – IV Estimation

	(1)	(2)	(3)	(4)
ln(slave exports / area)	-0.208***	-0.201***	-0.286*	-0.221***
	(0.045)	(0.049)	(0.133)	(0.076)
Colonizer FE	No	Yes	Yes	Yes
Geography Controls	No	No	Yes	Yes
Restricted Sample	No	No	No	Yes
Observations	52	52	52	42
First-stage F-statistic	5.18	5.02	0.99	1.70
Endogeneity test (p-value)	0.025	0.013	0.022	0.009
Sargan test (p-value)	0.177	0.298	0.652	0.509

Note: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.10. Slave exports normalized by land area. Instruments are sailing distances to Atlantic and Indian Ocean slave markets, and overland distances to trans-Saharan and Red Sea slave markets (Nunn 2008, pp. 160-161). ´

Our replication shows that the IV coefficients are approximately twice those from OLS, suggesting potential attenuation bias and consistent with Nunn's (2008) initial findings. While the results show slight variation in the F statistics relative to Table IV, all first-stage F statistics are below the commonly used threshold of 10 for strong instrumental variables (Stock and Yogo, 2005), which could undermine the model's confidence in explaining variance across the sample.

Furthermore, the exclusion criterion for distance in the analysis is that distance to slave markets would affect only the current GDP through slave exports. This assumption remains contentious, as distance in economic geography represents an innate condition for trade costs, the diffusion of technologies, and access to global markets, all of which have a considerable impact on GDP (Redding & Venables, 2004). The role of geography also stands as a potential measurement error, as GDP per capita remains a national average, whereas the slave trade, or diffusion of resources, remains a contextual and geographically bound phenomenon, which is not uniformly conditioned across nations, with a further consideration of whether slave trade works through other key determinants to GDP, rather than having a direct effect. This spatial mismatch could introduce measurement error while obscuring within-country heterogeneity (Whatley and Gillezeau 2011).

EXTENSION: Mediation Analysis

In relation to Africa and the slave trade, Nunn (2008) discusses two key parameters: ethnic fractionalization and state capacity (pp. 166-7), both of which show a bi-directional correlation with slave exports. While acknowledging that the correlation is estimated, this exercise seeks to decompose the potential total, direct, and indirect mediation effects of these channels on the effect of slave exports on log real GDP per capita. We have therefore employed the Beron and Kenny (1986) mediation framework, which decomposes the total effect into direct and indirect components. We examine two mediators: (1) ethnic fractionalization (Alesina et al. 2003) and (2) state development (the proportion of population with centralized pre-colonial institutions; Gennaioli and Rainer 2006, as cited in Nunn 2008, p. 165). All regressions include colonizer fixed effects and full geography controls to match Nunn's specifications.

Table 3: Channels of Causality: Total and Direct Effects Between Slavery, Ethnic Fractionalization, State Development and Direct Effects

	(1) Total Effect	(2) → Ethnic Frac.	(3) → State Dev.	(4) Direct Effect
ln(slave exports / area)	-0.076**	0.030***	0.015	-0.082*
	(0.029)	(0.008)	(0.015)	(0.045)
colony1	0.839	-0.079	0.107	0.591
	(0.552)	(0.150)	(0.222)	(0.530)
colony2	0.765	-0.078	0.100	0.603
	(0.560)	(0.152)	(0.226)	(0.535)
colony3	0.449	-0.078	0.114	0.680
	(0.592)	(0.168)	(0.261)	(0.617)
colony4	0.082	-0.303	0.657**	-0.174
	(0.770)	(0.209)	(0.309)	(0.782)
colony5	2.137**	-0.427*	0.272	1.884**
	(0.839)	(0.227)	(0.339)	(0.848)
colony6	1.341	0.272	-0.147	1.122
	(0.928)	(0.250)	(0.369)	(0.886)
colony7	0.752	0.300	0.122	0.999
	(0.833)	(0.227)	(0.334)	(0.841)
abs_latitude	0.016	-0.012**	0.020**	-0.012
	(0.017)	(0.005)	(0.008)	(0.021)
longitude	0.000	0.001	0.002	-0.009
	(0.005)	(0.001)	(0.003)	(0.006)
rain_min	-0.001	-0.004**	0.007**	-0.005
	(0.007)	(0.002)	(0.003)	(0.008)
humid_max	0.009	0.004	-0.004	0.006
	(0.012)	(0.003)	(0.005)	(0.011)
low_temp	-0.019	-0.001	-0.011	-0.036
	(0.028)	(0.008)	(0.012)	(0.029)

ln_coastline_area	0.085**	-0.007	-0.000	0.059
	(0.039)	(0.011)	(0.017)	(0.041)
ethnic_fractionalization				-0.366
				(0.770)
state_dev				0.174
				(0.499)
cons	5.944***	0.545*	0.401	7.185***
	(1.108)	(0.300)	(0.469)	(1.275)
Observations	52	51	47	47
R-squared	0.601	0.691	0.631	0.623

Note: Columns show: (1) Total effect of slave exports on GDP, (2) Effect of slave exports on ethnic fractionalization (3) Effect of slave exports on state development, (4) Direct effect after controlling for both mediators.
Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01

Our findings suggest that slavery indeed has a significant effect on current GDP per capita ($\beta = -0.076$, $p < .05$) and on ethnic fractionalization in our sample ($\beta = 0.030$, $p < 0.01$), even when controlling for colonial binary and geography. However, we do not find a relationship between slave exports and state development in the 19th century, contrary to Nunn (2008) 's Figure VII when controls are included. Our mediation analysis does not find any mediation between the effect of slave exports and GDP per capita through state development or ethnic fractionalization.

Our quantification advances Nunn's (2008) work in three ways. First, it confirms that Nunn (2008) has provided an account of the direct effect of slavery exports on GDP per capita among African nations, and we have not observed any statistical mediation between ethnic fractionalization and state development in our exercise. We do, however, reveal that ethnic fragmentation is potentially affected by slave exports. While mediation nor effect of ethnic fractionalization on GDP per capita has been found when accounting for Nunn (2008) control variables, Nunn and Wantchekon (2011) have identified fractionalization as a foundation of reduced trust, which relation to human capital destruction, demographic collapse, and institutional degradation, which could represent a multifaceted relationship between fractionalization, state development, and, GDP per capita.

The link between ethnic fractionalization has been further posited to reduce public goods provision (Alesina et al. 1999; Miguel and Gugerty 2005) and to lower economic growth (Easterly and Levine 1997). In addition to the persistence of colonial legacies, our findings shed light on the underlying mechanisms through which the impacts of the slave trade function through what Rodrik et al. (2004) refer to as *deep determinants*, a foundation for path-dependent development trajectories. The implications of which do not undermine Nunn's (2008) findings but rather point to a potentially multifaceted and systemic relational pattern between our sample size, chosen variables, and the conceptual complexity of the roots to economic development.

Unlike capital accumulation or human capital development, which represent tangible, quantifiable measures for governments to directly invest in, ethnic fragmentation and broader institutional systems represent foundations sustained by multiple reinforcing mechanisms of which low-hanging fruit are ripe, whereas systemic issues remain entrenched, embedded, and hard to shift. To that end, the policy implications of the structural issues imposed by ethnic fragmentation require power-sharing institutions, such as proportional representation, and broader social cohesion mechanisms to overcome colonial-era divisions (Posner 2004).

While our analysis has provided a basis for estimating the historical legacies that shape concurrent development outcomes, there are methodological limitations that warrant mention. For instance, the mediation analysis, or the respective work of Nunn (2008), is limited by a small sample size, which reduces the possibility of testing mediation significance, such as through a Sobel test, which generally requires a larger sample (Preacher & Leonardelli, 2001). Furthermore, the analysis does not capture within-country or longitudinal changes, which could reflect what Austin (2008) refers to as the *compression of history*, a dynamic that could undermine the evolutionary process of economic history by simplifying 500 years of complex historical relations.

CONCLUSION

This paper replicates Nunn's (2008) demonstration of the historical legacies of slave exports, which represents a statistically significant negative effect on the concurrent GDP per capita of African nations. Our analysis confirms the robust negative relationships in OLS and IV estimation and extends the initial analysis by conducting a mediation analysis of the relationship between slavery and GDP per capita via ethnic fractionalization and state development, which shows no statistically significant mediation effect. Nevertheless, slave export has been shown to significantly affect ethnic fractionalization. While a mediation effect was not found, our broader literature review suggests that a plausible counter-theory holds that ethnic fractionalization affects economic development by reducing the provision of public services, hindering human capital development, and facilitating low-trust environments. While the mediation analysis found no statistical significance, the research acknowledges that precise quantification of mediation channels requires a larger sample that allows for stronger identification strategies. The historical legacies of the conditions that shape development outcomes or entrench trajectories, however, warrant an approach that seeks to decompose the underlying, endogenous, and multicausal mechanisms that drive the long-term evolutionary process of economic change, as was sought to be presented through our work, and will hopefully be further shaped by economic historians in the future.

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